Calibrating What Behavior Can Identify: Conformal Reward Sets for Inverse Reinforcement Learning

Yuhan Chi
School of Mathematical Sciences
Fudan University
yhchi25@m.fudan.edu.cn

Abstract

Inverse reinforcement learning (IRL) returns a reward, and a policy is then optimized against it as if it were correct. Conformal inverse optimization (CIO) offers a principled alternative for one-shot inverse problems: calibrate a radius around the point estimate on held-out decisions, then optimize robustly over the resulting set. We ask what survives the move to IRL. The conformal step transfers untouched; the score and its downstream interpretation require care under reward non-identifiability. First, CIO scores a demonstration by an angle between reward vectors, which depends on coordinates the problem never fixes: a feature reparameterization that changes no reward and no demonstrator's policy still moves the calibrated angular radius by 24° on average at condition number 200. Scoring instead by the spread a reward difference induces over achievable policy values restores invariance and keeps the score a linear program. Second, building on CIO's intersection guarantee and inverse-set diameter assumption, we distinguish the events certified by different distances to the cone of rewards explaining new behavior: the nearest point gives intersection with that cone, the farthest point of the normalized cone gives containment of the latent reward. The first admits a general LP formulation, and at a 0.80 target we measure the two events at 0.82 and 0.44. We give the enlarged radius that prices the relaxation, and the point past which the resulting robust program has zero worst-case advantage over its reference. This threshold concerns our ball-based prescription, not an impossibility for all decisions from behavior.

1 Introduction

IRL turns observed behavior into a reward estimate, and that estimate is then optimized to pick a policy.[1] Uncertainty is dropped exactly where it matters, because two rewards that explain the demonstrations equally well can recommend different actions. Conformal inverse optimization [Lin et al., 2024] is an appealing remedy for the analogous one-shot problem: observe a population of decision makers, fit a point estimate on part of it, and on the rest record how far that estimate sits from the objectives that would rationalize each held-out decision. The conformal quantile of those distances defines an uncertainty set with a finite-sample guarantee, and a robust optimization over that set replaces the point prescription. The recipe is modular and looks portable. This paper reports what happens when it is carried into IRL, component by component (Table 1).

[1] Code: `https://github.com/Chi-Shan0707/cp-irl`. This author revision (6 October 2026) clarifies attribution and scope and adds Appendix D after peer review. The accepted submission and revision history are available in the repository.

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Table 1: The CIO recipe in IRL. The quantile, intersection event, and use of an inverse-set diameter bound are inherited; bold rows specify our adaptations.

CIO ingredient	Counterpart here	Status
Point estimate from decision data	any IRL fit (MaxEnt, Bayesian, ...)	inherited
Inverse-feasible set of a decision	Bellman cone $K(\pi)$, linear in θ	inherited
Score: angle to the estimate	span distance to $K(\pi)$; invariant, still an LP	**redesigned** (§4.2)
Split-conformal quantile	unchanged	inherited
Intersection + inverse-set diameter bound	unit-span containment radius $\min\{2, 2q_\gamma + \eta\}$	**geometry adapted** (§4.3)
Robust decision	reference-relative robust MDP, one LP	**redesigned** (§4.4)

One component transfers for free. Once per-demonstrator scores exist, the calibration arithmetic—rank them, take the $\lceil \gamma(N+1) \rceil$-th, invoke exchangeability—is split conformal prediction exactly as in Vovk et al. [2005] and Lin et al. [2024]; we inherit it wholesale. Sequential decision making changes nothing about that argument. The work is in the two components on either side of it: which distance gets calibrated, and what the resulting event is good for.

The score is measured in the wrong space. CIO scores a demonstration by the angle between the point estimate and the nearest objective that rationalizes it. In IRL that angle lives in coordinates the problem does not fix. A reward vector can change without changing the difference between any two policies' values: add a potential-shaping term, or alter the reward on a state no policy reaches. The angle registers such a change; behavior cannot. An uncertainty set built from angles can therefore hit its target rate while much of its radius covers differences no demonstration could exhibit. This is not hypothetical: we apply an invertible feature reparameterization that leaves every scalar reward and every demonstrator's policy pointwise unchanged, and the calibrated angular radius still moves by 24° on average (§5). Two identical problems get two different uncertainty sets. The fix is not a better estimator but a prior question: what does behavior actually distinguish? Everything observable here factors through the achievable occupancy measures, so we measure a reward difference v by the spread of policy values it produces, $\|v\|_{\mathcal{D}} = \max_\mu V_v(\mu) - \min_\mu V_v(\mu)$, a seminorm whose null space is exactly the directions behavior cannot see.

Intersection is not containment. This distinction is already explicit in CIO's intersection guarantee; its Assumption 2 bounds the Euclidean diameter of the inverse-feasible set and its Theorem 3 uses that bound for decision quality [Lin et al., 2024]. Here we express the corresponding identification requirement in normalized occupancy-span geometry. Split conformal certifies that *some* reward in the calibrated set makes the new demonstrator optimal; our route to reference-relative safety requires the demonstrator's *own* normalized reward to lie in the set. These events need not be close in either inverse optimization or IRL. At target 0.8 we measure the first event at 0.82 and the second at 0.44, and no amount of calibration closes that gap. The two events differ by which distance to the cone is calibrated. Scoring the distance to the *nearest* point—the quantity a linear program can compute—certifies intersection. Scoring the distance to the *farthest* point of the normalized cone certifies containment for nonnull latent rewards without a width assumption, but is generally a nonconvex maximization. Our construction is the convex relaxation together with an explicit account of what it costs: a factor of two from renormalizing the witness, plus a certified bound η on the cone's width, giving $R = \min\{2, 2q_\gamma + \eta\}$. Using only the universal bound $\eta = 2$ makes this ball-based prescription degenerate. A sharper certificate may follow from the known geometry itself or from additional information; Appendix D supplies a small exact example.

What we report is diagnostic rather than a performance improvement: half of CIO's pipeline moves to IRL untouched, two components must be rebuilt in decision space, and the step from intersection to containment is a relaxation whose price we state rather than hide. Regret against point-estimate IRL is therefore not the yardstick, and §5 reports where our construction does and does not lower it.

2 Related work

This work sits between two literatures. Inverse optimization recovers objectives from observed decisions [Ahuja and Orlin, 2001, Chan et al., 2019], and CIO adds finite-sample calibration to it [Lin et al., 2024]; this is a different problem from data-driven robust optimization, where the coefficients are observed or predicted rather than latent [Chenreddy et al., 2022, Sun et al., 2023]. IRL has known since Ng and Russell [2000] that behavior determines a reward only up to a cone, and recent work characterizes those feasible sets and asks what extra structure shrinks them—experts of known suboptimality under a generative model [Poiani et al., 2024], several demonstrators of declared suboptimality [Kim et al., 2026], a unified LP view of demonstrations and feedback [Kim et al., 2024], or an account of what behavior provably cannot resolve [Skalse et al., 2023] and what that costs a downstream decision [Lazzati and Metelli, 2025]. That line attaches no predictive coverage guarantee for a new demonstrator to its set. CIO already treats set identification through intersection coverage and a diameter assumption. Our adaptation uses the IRL occupancy geometry to obtain an invariant LP score, a normalized containment bridge, and a precise degeneracy threshold for the resulting reference-relative prescription.

For the geometry we borrow rather than reinvent. EPIC canonicalizes shaping away [Gleave et al., 2021], DARD restricts to realizable transitions [Wulfe et al., 2022], and STARC generalizes the canonicalize-then-normalize recipe, showing that any pseudometric with worst-case-regret upper *and* lower bounds is bilipschitz equivalent to a STARC metric [Skalse et al., 2024]. Our occupancy span plays the role of a canonicalization–normalization pair in the same spirit, taken relative to the known P, so we propose no new reward metric; what is new is using this geometry as a conformal score over Bellman cones, and being explicit about the identification assumption the resulting radius still needs. On the robust side, tractability is classically recovered by rectangularity [Iyengar, 2005, Nilim and El Ghaoui, 2005, Wiesemann et al., 2013], and Gadot et al. [2024] show that non-rectangular, ball-shaped reward ambiguity is tractable in occupancy space—the space we also work in, with the difference that their ambiguity set is given rather than calibrated. Conformal methods have appeared in RL and decision making for other purposes [Gupta et al., 2023, Pan et al., 2026, Zhou and Zhu, 2025], but we are not aware of prior work that calibrates an inverse-feasible reward set from a population of IRL demonstrators.

3 Setup

A tabular MDP $M = (S, A, P, \rho_0, \beta, \phi)$ has known kernel P, initial distribution ρ_0, discount $\beta \in [0, 1)$, and features $\phi : S \times A \to \mathbb{R}^d$ inducing $r_\theta(s, a) = \phi(s, a)^\top \theta$. A population of demonstrators draws rewards $\hat{\theta}_1, \ldots, \hat{\theta}_N \sim \Lambda$ i.i.d.; demonstrator k acts optimally under $\hat{\theta}_k$, producing a deterministic policy π_k that we assume is observed exactly rather than estimated from finite trajectories. Let $M(P)$ be the polytope of normalized discounted occupancy measures, Φ the $(|S||A|) \times d$ feature matrix, and $V_\theta(\mu) := \theta^\top \Phi^\top \mu/(1 - \beta)$. The rewards that make an observed policy optimal form the cone

$$K(\pi) := \{\theta : Q^\pi_\theta(s, a) \le Q^\pi_\theta(s, \pi(s)) \; \forall s, a\}.$$

Substituting the Bellman resolvent removes the unknown value function and makes $K(\pi)$ exactly $|S|(|A| - 1)$ halfspaces in θ alone (Proposition 5), which is the sequential restatement of Ng and Russell's feasible-reward characterization.

Assumption 1 (Exchangeable calibration unit)**.** *The pairs $(\hat{\theta}_k, \pi_k)$, $k = 1, \ldots, N_{\mathrm{val}}, N_{\mathrm{val}}+1$, are i.i.d. (hence exchangeable), and M—in particular P—is shared and fixed across all demonstrators and the deployment instance.*

Each demonstrator therefore plays the role of one (decision, context) pair in CIO's setting; per-trajectory exchangeability within a single demonstrator would instead be broken by within-trajectory Markov dependence. The pipeline is Table 1 read top to bottom: split the population, fit $\bar{\theta}$ on the training split with any IRL method, calibrate on the rest—write N_{val} for the size of that split—convert intersection into containment, and solve a robust MDP. The inherited CIO baseline uses the spherical cap $C(\bar{\theta}, \alpha) = \{\theta : \|\theta\|_2 \le 1, \; \theta^\top \bar{\theta} \ge \cos \alpha\}$ (Appendix A); that specific choice, not the calibrate-then-robustify idea, is what reward non-identifiability damages.

4 Calibrating in decision space

4.1 A distance that behavior can see

For $v \in \mathbb{R}^d$, define

$$\|v\|_{\mathcal{D}} := \max_{\mu \in M(P)} V_v(\mu) - \min_{\mu \in M(P)} V_v(\mu) = V_v^*(\rho_0) + V_{-v}^*(\rho_0) = h_{D_F}(v),$$

the spread of policy values that v produces, i.e. the support function h_{D_F} of the feature difference body $D_F := \Phi^\top(M(P) - M(P))/(1-\beta)$, a form we return to in §4.4. The right-hand form is two value iterations; no linear program is needed for the seminorm itself.

Proposition 1 (Properties of $\|\cdot\|_{\mathcal{D}}$)**.** *$\|\cdot\|_{\mathcal{D}}$ is a seminorm on $\mathbb{R}^d$, and $\|v\|_{\mathcal{D}} = 0$ exactly when $v \in \mathcal{N} := \{a : a^\top \Phi^\top(\mu - \nu) = 0 \ \forall \mu, \nu \in M(P)\}$, the subspace of directions that add the same constant to every policy's value. $\mathcal{N}$ contains the potential-shaping subspace and can be strictly larger. For any θ, θ', with μ_θ^* optimal for θ and $\mu_{\theta'}$ optimal for θ',*

$$\mathrm{Reg}_\theta(\pi_{\theta'}) := V_\theta(\mu_\theta^*) - V_\theta(\mu_{\theta'}) \ \le \ \|\theta - \theta'\|_{\mathcal{D}}.$$

Finally $\|\cdot\|_{\mathcal{D}}$ is exactly invariant under $\phi \mapsto A\phi$, $\theta \mapsto A^{-\top}\theta$ for any invertible A.

The regret constant is 1, with no Cauchy–Schwarz slack and no feature-norm Lipschitz factor, because the seminorm is defined as the value spread it has to bound. No seminorm that bounds regret can be smaller.

Proposition 2 (Minimality)**.** *Let N be any seminorm on $\mathbb{R}^d$ with $\mathrm{Reg}_\theta(\pi_{\theta'}) \le N(\theta - \theta')$ for all θ, θ'. Then $N(v) \ge \|v\|_{\mathcal{D}}$ for every v.*

Proof. Fix v and $\delta > 0$, and take $\theta = v$, $\theta' = -\delta v$. Any policy optimal for θ' maximizes $V_{-\delta v}$ and so minimizes V_v, while μ_θ^* maximizes V_v; hence $\mathrm{Reg}_\theta(\pi_{\theta'}) = \max_\mu V_v - \min_\mu V_v = \|v\|_{\mathcal{D}}$. The hypothesis with $\theta - \theta' = (1+\delta)v$ and homogeneity of N give $\|v\|_{\mathcal{D}} \le N((1+\delta)v) = (1+\delta)N(v)$. Let $\delta \downarrow 0$. □

So $\|\cdot\|_{\mathcal{D}}$ is the pointwise smallest seminorm that bounds worst-case regret: any valid alternative is at least as large, and a strictly larger one certifies a strictly weaker statement at the same radius. The restriction to $\delta > 0$ is not cosmetic—at $\delta = 0$ every policy is optimal for $\theta' = 0$ and the construction says nothing. Since shaping directions lie in $\mathcal{N}$, replacing θ by $\bar{\theta} + c$ for $c \in \mathcal{N}$ leaves both the score and the calibrated set unchanged; the angular score, computed from $\bar{\theta}$'s normalized coordinates, has no such property. Optimal behavior cannot identify positive reward scale either, and we remove that symmetry without building quotient coordinates: for any behaviorally nonnull reward θ we set $\theta^\circ := \theta/\|\theta\|_{\mathcal{D}}$, and compare rewards by $\|\theta^\circ - \theta'^\circ\|_{\mathcal{D}}$, a pseudometric with range $[0, 2]$ that vanishes exactly when $\theta' = c\theta + a$ for some $c > 0$, $a \in \mathcal{N}$, and that inherits the normalized regret bound (Proposition 6).

4.2 Score, quantile, and what they certify

Write $\bar{\theta}^\circ = \bar{\theta}/\|\bar{\theta}\|_{\mathcal{D}}$ for the *unit-span center* and define the *span score* s_k and *span ball* B_q,

$$s_k := \min_{\theta \in K(\pi_k)} \|\theta - \bar{\theta}^\circ\|_{\mathcal{D}}, \qquad B_q := \{\theta : \|\theta - \bar{\theta}^\circ\|_{\mathcal{D}} \le q\}.$$

Throughout, *coverage* names only the property $\Pr[E] \ge \gamma$ for a stated event E. The two events at issue are *intersection*, $K(\pi_{\mathrm{new}}) \cap B \ne \emptyset$, and *containment*, $\hat{\theta}^\circ_{\mathrm{new}} \in B$. Split conformal delivers coverage of whichever event the score's sublevel set defines, and nothing more.

Because $\|\cdot\|_{\mathcal{D}}$ has a linear-program epigraph and $K(\pi_k)$ is polyhedral, s_k is a single LP in $d + 2|S|$ variables, with no second-order cone anywhere; this is what would be lost by intersecting the cone with a Euclidean ball. Since $0 \in K(\pi_k)$ and $\|\bar{\theta}^\circ\|_{\mathcal{D}} = 1$, we have $s_k \in [0, 1]$, and both the score and its quantile are dimensionless and unchanged under $\bar{\theta} \mapsto c\bar{\theta} + a$ for $c > 0$, $a \in \mathcal{N}$.

Proposition 3 (Intersection coverage)**.** *Under Assumption 1, let q_γ be the τ-th smallest s_k among N_{val} calibration demonstrators, with $\tau = \lceil \gamma(N_{\mathrm{val}} + 1) \rceil$ and the sentinel $+\infty$ if $\tau = N_{\mathrm{val}} + 1$. For a fresh i.i.d. demonstrator, $s_{\mathrm{new}} \le q_\gamma$ holds if and only if $K(\pi_{\mathrm{new}}) \cap B_{q_\gamma} \ne \emptyset$, and $\Pr[K(\pi_{\mathrm{new}}) \cap B_{q_\gamma} \ne \emptyset] \ge \gamma$.*

The probability is the standard split-conformal rank argument [Vovk et al., 2005] applied verbatim. What belongs to this construction is the equivalence, immediate from s_k being a minimum: coverage of the score *is* intersection coverage of the ball, with no lossy conversion in between. The contrast is the obvious repair of the angular cap, which calibrates the value gap under $\bar{\theta}$ and converts it to a Euclidean ball by inverting a Lipschitz bound (Appendix A). That bound licenses gap $\leq$ const $\times$ distance and never the reverse, so its radius certifies neither event—a measured containment rate of 0.004 in Figure 1(b).

4.3 From intersection to containment

$K(\pi)$ is not an equivalence class. Even after removing scale and null directions it can contain rewards with genuinely different policy orderings, so touching the cone says little about where the demonstrator's reward is. Write $F^\circ(\pi) := \{\theta \in K(\pi) : \|\theta\|_{\mathcal{D}} = 1\}$ for the normalized cone, understood modulo the null directions.

Which event the quantile certifies depends only on which distance to $F^\circ(\pi)$ is scored. The latent reward lies in $F^\circ(\pi_{\mathrm{new}})$, so scoring the *largest* distance,

$$s^{\max}(\pi) := \max_{u \in F^\circ(\pi)} \|u - \bar{\theta}^\circ\|_{\mathcal{D}},$$

gives $\|\hat{\theta}^\circ_{\mathrm{new}} - \bar{\theta}^\circ\|_{\mathcal{D}} \leq s^{\max}(\pi_{\mathrm{new}})$ pointwise, and the same split-conformal argument then certifies containment directly, under Assumption 1 and nonnull latent rewards, without a cone-width certificate. This is the score we want. It is also the maximization of a convex function over a generally nonconvex set. We do not solve the general problem; Appendix D solves a special case exactly. Our s_k scores the *smallest* distance instead: a linear program, but it bounds the wrong side. One assumption converts one into the other.

Assumption 2 (Certified cone width)**.** *There is a known $\eta \in [0, 2]$ with $\sup_{u,v \in F^\circ(\pi)} \|u - v\|_{\mathcal{D}} \leq \eta$ almost surely for the deployment population, and each latent reward is behaviorally nonnull.*

Under Assumption 2, $s^{\max}(\pi) \leq 2s(\pi) + \eta$ pointwise: when $s(\pi) < 1$ the near witness renormalizes at a cost of $2s(\pi)$ and the cone width covers the rest, and when $s(\pi) = 1$ the bound holds trivially because $s^{\max} \leq 2$. Since $t \mapsto 2t + \eta$ is increasing it commutes with order statistics, so the pointwise inequality passes straight to the quantile, $q^{\max}_\gamma \leq 2q_\gamma + \eta =: R$, which is what Theorem 1 uses. Read this way, Assumption 2 is additional information needed by this LP-to-containment bound, not by exact maximum-distance calibration. The factor 2 renormalizes the witness, and η stands in for what an exact $s^{\max}$ would have measured. A small η can be certified from a specific known MDP, as in Appendix D. Side information, multiple environments, or active comparisons can instead restrict the feasible set; that route requires reformulating the score and certificate for the restricted set. Without a sharper certificate we use the universal bound $\eta = 2$.

Theorem 1 (Two-stage containment and safe prescription)**.** *Under Assumptions 1–2, let q_γ be the Stage-I quantile and set $R := \min\{2, 2q_\gamma + \eta\}$. Then $\Pr[\|\hat{\theta}^\circ_{\mathrm{new}} - \bar{\theta}^\circ\|_{\mathcal{D}} \leq R] \geq \gamma$. Let $\mu_R \in \arg\max_{\mu \in M(P)} \min_{\theta \in B_R} \big[V_\theta(\mu) - V_\theta(\mu_{\mathrm{ref}})\big]$ for a fixed reference $\mu_{\mathrm{ref}} \in M(P)$; then*

$$\Pr\Big[V_{\hat{\theta}_{\mathrm{new}}}(\mu_R) \geq V_{\hat{\theta}_{\mathrm{new}}}(\mu_{\mathrm{ref}})\Big] \geq \gamma.$$

The argument is short. On the Stage-I event some $\tilde{\theta} \in K(\pi_{\mathrm{new}})$ is within q_γ of the unit-span center, so if $q_\gamma < 1$ the reverse triangle inequality makes $\tilde{\theta}$ nonnull; normalizing it moves it by at most another q_γ, which is where the factor 2 comes from, and Assumption 2 then covers the remaining distance to the latent reward inside the same cone. Minimizing over $F^\circ(\pi)$ instead of $K(\pi)$ would remove the factor 2, giving $R = \min\{2, q_\gamma + \eta\}$; maximizing rather than minimizing over $F^\circ(\pi)$ removes η as well. Both are generally nonconvex, and both are what keeping Stage I a linear program costs.

4.4 The deployed policy

B_R is unbounded—it contains the whole affine subspace $\bar{\theta}^\circ + \mathcal{N}$—so worst-case *absolute* value over it is unbounded below in general. The reading we take is that absolute value is the wrong criterion: a direction with no behavioral content can be scaled without bound, and only a comparison

between two occupancy measures is insensitive to it. We therefore optimize worst-case value relative to a reference μ_{ref}. With $x := \Phi^\top(\mu - \mu_{\mathrm{ref}})/(1-\beta)$ and D_F the feature difference body $\Phi^\top(M(P) - M(P))/(1-\beta)$,

$$\min_{\theta \in B_R} \left[V_\theta(\mu) - V_\theta(\mu_{\mathrm{ref}})\right] = \left[V_{\bar{\theta}^\circ}(\mu) - V_{\bar{\theta}^\circ}(\mu_{\mathrm{ref}})\right] - R\,\gamma_{D_F}(x),$$

where γ_{D_F} is the Minkowski gauge of D_F, the polar dual of $\|\cdot\|_{\mathcal{D}}$. Since $\mu - \mu_{\mathrm{ref}}$ ranges over $M(P) - M(P)$ by construction, the penalty is automatically normalized to $[0,1]$, R reads directly in regret units, and no sign condition can arise. The gauge has an LP epigraph, so the whole robust problem is one linear program.

Proposition 4 (Degeneracy threshold)**.** *For the unit-span center, if $R \geq 1$ the optimal relative robust value is zero, attained by μ_{ref}; if $R > 1$ every optimizer is feature-equivalent to the reference. A nontrivial safe improvement therefore requires $2q_\gamma + \eta < 1$.*

This is a necessary condition for strictly positive worst-case advantage for this ball and objective, not a sufficient condition for improvement and not an information-theoretic impossibility for other uncertainty sets or prescriptions.

Vacuity is therefore auditable: the implementation reports q_γ, η, R, and whether $R < 1$. One modelling choice remains. Centering at $\bar{\theta}$'s own optimal occupancy is degenerate, since the reference is then optimal for every R; all experiments instead center at the calibration population's mean occupancy, which asks whether a policy beats what a random draw from the population achieves. Since Theorem 1 guarantees dominance only against whichever reference is chosen, a sensitivity analysis over that choice is required before deployment.

5 Experiments

The original experiments E1–E3 are deliberately narrow, each testing one link in the calibrate–identify–prescribe chain: whether the calibrated radius survives an intervention that changes nothing observable (E1), whether intersection coverage implies containment (E2), and how strong an external certificate Stage II would need (E3). Each isolates one mechanism rather than benchmarking end-to-end regret. The separately labelled post-acceptance diagnostic in Appendix D exercises a nondegenerate certificate and a set-shape control; it does not change the E1–E3 results.

E2 and E3 use random tabular MDPs, a Voronoi-feature gridworld, and Objectworld [Levine et al., 2011], each with discount $\beta = 0.9$, $|S| \leq 36$, and $d \leq 4$. We fit MaxEnt IRL [Ziebart et al., 2008] and Bayesian IRL [Ramachandran and Amir, 2007] by Metropolis–Hastings on 20 training demonstrators, calibrate on 20 at target level $\gamma = 0.8$, and evaluate on 20 fresh demonstrators, for 10 environment seeds: 60 runs and 1,200 reward evaluations (600 distinct test draws, shared across estimators). Draws within a run share a fitted center and radius, so intervals are 95% Student-t intervals over the 30 environment–seed blocks after averaging the two estimators within each block. Computation used a 20-core laptop, no GPU, and no commercial solver. An exact-fit suboptimality LP baseline is not reported: it returned a span-zero center in one of 30 fits, for which unit-span normalization is undefined, and we record that as an abstention rather than perturbing the reward by an arbitrary epsilon.

E1: an intervention that changes nothing. For each $\kappa \in \{1, 2, 5, 10, 50, 200\}$ and each of 10 seeds we draw a random MDP ($|S| = 6$, $|A| = 3$, $d = 4$), a center $\bar{\theta}$, and 16 calibration demonstrators, together with a random symmetric positive-definite A of condition number κ, and calibrate before and after $\phi \mapsto A\phi$, $\theta \mapsto A^{-\top}\theta$; here the span score is taken around $\bar{\theta}$ itself, which Proposition 1 makes invariant as well. Coordinates change; every scalar reward $\phi(s,a)^\top\theta$ does not (maximum numerical discrepancy 2.1×10^{-14}), so a behaviorally meaningful radius should not move. At $\kappa = 200$ the angular radius changes by 0.424 radians ($24.3°$) on average and 1.071 radians in the worst seed, while the span radius changes by at most 2.0×10^{-8} (Figure 1a). This is a falsification test rather than a favorable data distribution: the compared problems are the same problem.

E2: intersection holds, containment does not. At $\gamma = 0.8$, pooled empirical intersection rates are 0.812 ± 0.042, 0.838 ± 0.046, and 0.819 ± 0.036 for the angular, value-gap, and span scores; the first and third are exactly inverse-set intersection, the second only score coverage. Containment at the

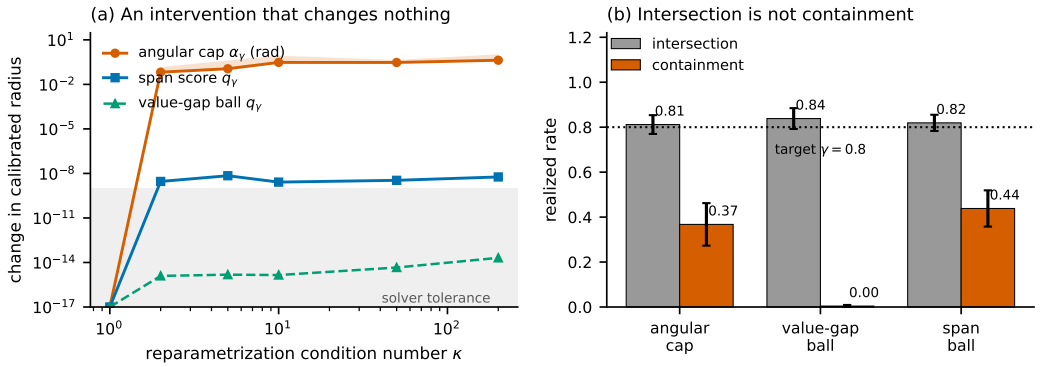


Figure 1: The two failures. **(a)** A random invertible feature reparameterization of condition number κ leaves every pointwise reward and policy unchanged, yet the inherited angular cap moves; the span score does not, up to solver tolerance. The value-gap score is also invariant—it is computed from values rather than from coordinates—and sits lower still because it needs no LP; §4.2 and Appendix A show that its converted radius nonetheless certifies neither event, so invariance alone is not sufficient. Shading spans mean to worst over 10 seeds. **(b)** At $\gamma = 0.8$ every construction attains score-event coverage at target: intersection for angular and span, value-gap coverage for the other. Containment at the same Stage-I radius falls short. The right-hand bars use each construction's own geometry (angular, Euclidean, and span, respectively). The value-gap conversion certifies neither set event. Error bars are 95% t intervals over 30 environment–seed blocks.

same Stage-I radius is only 0.368 ± 0.095 for the angular cap and 0.438 ± 0.081 for the span ball, while the value-gap-to-Euclidean conversion contains 0.004 ± 0.005 (Figure 1b). Conformal ranking behaves exactly as the theory says; containment does not follow from q_γ alone. The separation is a property of which distance is scored, not of the calibration: the same rank argument applied to $s^{\max}$ would place the second bar at the target as well.

E3: how much identification is missing? The calibrated q_γ has pooled mean 0.316, median 0.289, and range $[0.019, 0.871]$, so the observable budget $b := \max\{1 - 2q_\gamma, 0\}$ is usually positive but small. A certificate must satisfy $\eta < b$ to meet the necessary condition $R < 1$ for strictly positive worst-case advantage: with $\eta = 0$, 0.2, and 0.5, respectively 85.0%, 76.7%, and 36.7% of runs meet that condition, without thereby establishing improvement. The environment matters (at $\eta = 0.2$, 95% for random MDPs, 75% for Objectworld, 60% for gridworld). The universal certificate $\eta = 2$ gives $R = 2$ everywhere, with containment at radius R equal to 1.000, reference dominance 1.000, and nontriviality 0.000—a consistency check of the degenerate case, where the implementation explicitly returns the reference, not evidence of a useful safety improvement. The full frontier is in Figure 2.

What these experiments do not show. Using the Stage-I radius directly can lower or raise true-reward regret; in the balanced six-cell audit it lowers mean regret in two cells of six and raises it in four (Table 2). That policy is not covered by Theorem 1, which uses R, and our relative objective guarantees dominance over the reference rather than over the point-estimate policy. We do not report the scale, shift, and RLHF relabeling probes here: they are exploratory and outside the claims of Theorem 1.

6 Discussion

Metric choice is real and measurable (E1), but a conformal intersection radius alone does not certify containment of the latent reward (E2). Theorem 1 prices that gap and E3 makes the price observable: strictly positive worst-case advantage under this construction requires a certified cone width below the run-specific budget $1 - 2q_\gamma$. The universal width 2 proves containment but pushes the robust solution back to the reference: safe, but with no decision content. This diagnoses the chosen radius bound and ball-based prescription; it does not separate intrinsic non-identifiability from conservatism introduced by the bound or the set shape. The control in Appendix D illustrates that distinction. Two routes to a more informative construction are as follows.

The algorithmic route is to compute $s^{\max}$ itself, which would need no η at all. Since $\|v\|_{\mathcal{D}}$ is a maximum of finitely many linear functionals of v—one per pair of vertices of $M(P)$—its unit ball is polyhedral, so $F^{\circ}(\pi)$ is the intersection of a polyhedral cone with that boundary, a union of finitely many polyhedral pieces, and the maximum of a convex function over such a union is attained at a vertex of one of them. For small d this is enumerable; the complexity in general, and whether a useful relaxation sits between the LP and the exact program, we leave open. Appendix D implements an exactly enumerable case. The data route is to certify a small η: repeated observations of one demonstrator under several dynamics, active comparisons, or externally known restrictions on the reward. The two routes are complementary—one shrinks what must be assumed, the other strengthens what can be assumed—and either moves the same frontier in Figure 2.

Our exchangeability unit is demanding—a demonstrator with a fully observed exact optimal policy in a shared known tabular MDP—and finite trajectories, approximate optimality, subgroup-conditional coverage, and MDP shift each need separate analysis. Covariate-shift weighting [Tibshirani et al., 2019] may address the statistical part of transfer but not the identifiability conditions studied by Schlaginhaufen and Kamgarpour [2024].

The closest setting we have not tested is preference-based reward modelling [Christiano et al., 2017, Rafailov et al., 2023], where the same two questions recur with a different calibration unit: one annotator, one calibration point. A Bradley–Terry reward model is identified only up to the transformations that leave every modelled comparison unchanged, so the rewards consistent with an annotator's preferences again form a set rather than a point, and the distinction this paper is about—whether a calibrated radius merely meets that set or contains the annotator's own reward—applies unchanged. Two things do change. The comparison set is a polyhedral cone only when the comparisons are noiseless and the model is linear in features, so $K(\pi)$ must be replaced by whatever the preference likelihood makes feasible; and annotators are not exchangeable in the way Assumption 1 needs when they are recruited in batches or disagree systematically. We raise this because a reward model trained on preferences is then optimized against by a policy, which is structurally the pipeline we have been auditing, not because our experiments say anything about it. The pattern still generalizes past IRL: whenever an inverse problem is only partially identified, a calibrate-then-prescribe pipeline must answer three questions separately—in which geometry the score is measured, how the quantile is computed, and which event the calibrated set certifies about the latent parameter—and the conformal rank argument supplies the coverage step once the score and event are specified. CIO already states an intersection event and an inverse-set diameter assumption. Here their consequences are made explicit in IRL's normalized occupancy geometry; thin inverse sets are not assumed to be an automatic property of one-shot problems.

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A Deferred constructions

Proposition 5 (Resolvent-based score reduction)**.** *Fix π; let $P_\pi(s,s') = P(s,\pi(s),s')$, let Φ_π be the feature matrix restricted to π's actions, and let $W_\pi := (I-\beta P_\pi)^{-1}\Phi_\pi$. Then $V_\theta^\pi = W_\pi\theta$ and $Q_\theta^\pi(s,a) = \Psi_\pi(s,a)^\top\theta$ with $\Psi_\pi(s,a) := \phi(s,a) + \beta P(s,a,\cdot)W_\pi$, both linear in θ, so*

$$K(\pi) = \{\theta :\ \Psi_\pi(s,a)^\top\theta \le \Psi_\pi(s,\pi(s))^\top\theta\ \forall s,a\}$$

exactly. The angular baseline additionally uses $\hat\Theta_2(\pi) = K(\pi)\cap\{\theta : \|\theta\|_2 \le 1\}$. Both have d decision variables instead of $d+|S|$ and $|S|(|A|-1)$ nontrivial linear inequalities, after one resolvent solve per distinct policy.

We verify this against a joint-(θ, V) reference to solver precision, at 2–5.5$\times$ the speed for $|S|$ from 20 to 100, and update it across demonstrators by Sherman–Morrison–Woodbury in $O(k|S|^2 + k^3)$ when policies differ at k states.

Proposition 6 (Projective span geometry)**.** *$d_\circ(\theta,\theta') := \|\theta^\circ - \theta'^\circ\|_{\mathcal{D}}$ is a pseudometric with range $[0,2]$, invariant to independent positive rescaling, addition of elements of $\mathcal{N}$, and invertible feature reparameterization. For nonnull rewards $d_\circ(\theta,\theta') = 0$ iff $\theta' = c\theta + a$ for some $c>0$ and $a\in\mathcal{N}$, and $\mathrm{Reg}_\theta(\pi_{\theta'})/\|\theta\|_{\mathcal{D}} \le d_\circ(\theta,\theta')$.*

Theorem 2 (Concavity under reward-only ambiguity; failure with dynamics ambiguity)**.** ***(a)** For* any *nonempty compact $C\subseteq\mathbb{R}^d$, if P is fixed and known then $\max_{\mu\in M(P)}\min_{\theta\in C}\theta^\top\Phi^\top\mu$ is a concave maximization over the fixed polytope $M(P)$, hence a convex outer problem whenever C admits a tractable linear-minimization oracle. **(b)** The guarantee need not survive when ambiguity couples reward and transition: there is a 2-state, 2-action MDP and a convex one-dimensional non-product ambiguity set over (θ,P) whose worst-case value function is not concave.*

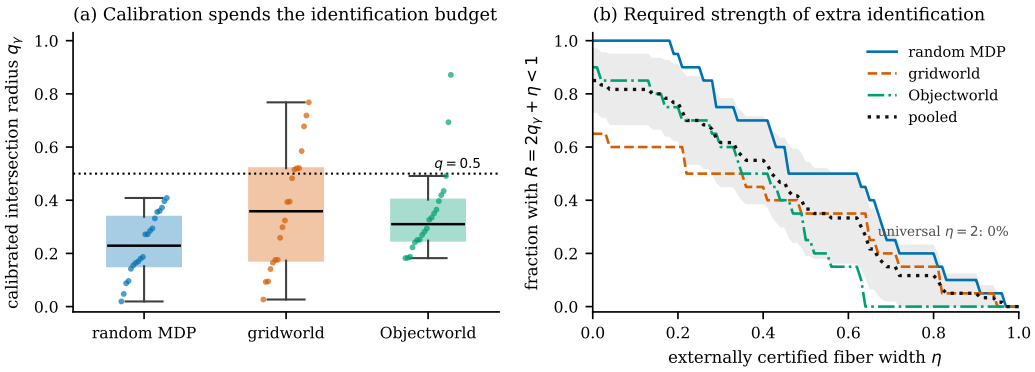


Figure 2: Identification-budget frontier. **(a)** Distribution of the dimensionless Stage-I radius over 20 runs per environment (two estimators and 10 seeds); even a zero-width cone can be nontrivial only below $q_\gamma = 0.5$. **(b)** For each hypothetical certified width η, the fraction of runs satisfying $2q_\gamma + \eta < 1$. The curve does not estimate or validate η; it states how strong an independent identification mechanism must be. Gray shading is a 95% t interval for the pooled fraction over 30 environment–seed blocks.

Part (a) is why the robust step is routine once written in occupancy space, and covers the fixed-kernel regime for an arbitrary compact set. Part (b) is existential (Appendix C.2); rectangular and other structured cases remain tractable, and failure of concavity is not a hardness proof. Xu et al. [2026] instead use d-rectangular sets whose reward and transition components vary together across data sources. Part (a) does not cover that setting, since fixing P removes exactly the coupling; what it does cover is the reward-only restriction of such a set, which is one instance of the compact C.

The inherited angular baseline. For this baseline only, we replace every nonzero point estimate by its Euclidean normalization, $\|\bar{\theta}\|_2 = 1$. With the score $c_k = \max_{\theta \in \hat{\Theta}_2(\pi_k)} \theta^\top \bar{\theta}$ and $\alpha_\gamma := \arccos(q_\gamma)$, split-conformal calibration on c_k is CIO's exchangeability argument transplanted verbatim [Vovk et al., 2005], giving $\mathbb{P}(\hat{\Theta}_{2,\text{new}} \cap C(\bar{\theta}, \alpha_\gamma) \neq \emptyset) \geq \gamma$; we omit the routine restatement. This is intersection nonemptiness, not containment. Because the zero reward is Bellman-feasible for every policy, any cap with $\alpha \geq \pi/2$ makes the intersection event automatic, so intersection coverage alone never establishes reward identification. Under the sign condition $a(x) \geq -\|x\|_2 \cos\alpha$ (writing $x = a(x)\bar{\theta} + x_\perp$), the support function of the cap reduces to $a(x)\cos\alpha - b(x)\sin\alpha$ with $b(x) = \|x_\perp\|_2$, so the robust policy solves the point estimate with a fixed-weight ℓ_2 penalty $\tan\alpha_\gamma \cdot b(\Phi^\top \mu)$, set by the conformal quantile rather than tuned.

The value-gap repair and why it is incomplete. Replacing the angular score by the suboptimality gap under $\bar{\theta}$, $c_k := V_{\bar{\theta}}(\mu^*_{\bar{\theta}}) - V_{\bar{\theta}}(\mu_{\pi_k})$, removes shaping contamination for free: shaping directions earn a constant on every $\mu \in M(P)$ (an immediate consequence of the flow-conservation constraints defining $M(P)$), so adding one to $\bar{\theta}$ leaves every c_k unchanged. But converting this score to a Euclidean ball of radius $\rho := q_\gamma/\bar{\nu}$ needs a Lipschitz constant $\bar{\nu} := \|\phi\|_{2,\infty}/(1-\beta)$, and the bound licenses gap $\leq$ const $\times$ distance and never the reverse, so q_γ alone certifies nothing about the set (measured containment 0.004, Figure 1b). Its robust program is at least unconditional—$\min_{\|\theta-\bar{\theta}\|_2 \leq \rho} \theta^\top x = \bar{\theta}^\top x - \rho\|x\|_2$ by Cauchy–Schwarz—but an unconditional penalty on an uncertified radius is not a certificate.

B Additional experiments

Policy regret, for completeness. Table 2 reports mean true-reward regret in the balanced six-cell audit. The Stage-I span policy beats the point estimate in two of six cells and loses in four; the certified Stage-II policy at R is worse than the point estimate everywhere in this audit. Proposition 4 explains its return to the reference when $\eta = 2$, but does not predict the reference's regret relative to the point estimate. None of these comparisons is covered by Theorem 1, whose guarantee is

dominance over μ_{ref} and not over the point estimate; we report the table so that the claim in §5 has a source, not as evidence of a performance gain.

Table 2: Mean true-reward regret over 10 seeds per cell, at $\gamma = 0.8$. Lower is better; bold marks cells where the span policy beats the point estimate. *Span* uses the Stage-I radius q_γ and carries no containment guarantee; *safe* uses the certified R of Theorem 1 with $\eta = 2$.

Environment	Estimator	point	span	safe (R)	angular	value-gap
gridworld	Bayesian	0.019	0.033	0.051	0.027	0.016
gridworld	MaxEnt	0.038	**0.033**	0.051	0.046	0.031
Objectworld	Bayesian	0.028	**0.018**	0.035	0.079	0.028
Objectworld	MaxEnt	0.018	0.019	0.035	0.043	0.031
random MDP	Bayesian	0.003	0.009	0.034	0.049	0.004
random MDP	MaxEnt	0.004	0.008	0.034	0.025	0.005

Concentration baseline. Replacing the exact conformal quantile by a DKW bound shifts the quantile level by $\varepsilon = \sqrt{\log(2/\delta)/(2N)}$. On the angular score this never gave a narrower cap than conformal calibration in any case tested. On the value-gap score, whose threshold is an upper quantile, the corrected level $\gamma + \varepsilon$ exceeds 1 at $\gamma = 0.8$ and $\delta = 0.05$ whenever $N \leq 46$, so at our calibration sizes the DKW radius is infinite and certifies nothing. Capping it at an arbitrary $10\|\bar{\theta}\|_2$ gave true-reward regret 4–460$\times$ that of the point estimate, a figure that reflects the cap rather than the bound.

Reproduction. From the repository root, the main artifacts are generated by `experiments/run_reparam_invariance.py`, `experiments/run_containment_audit.py`, and `experiments/run_center_validity.py`; the figures are generated by `paper/make_fig1_metric.py` and `paper/make_fig2_identification_frontier.py`. The concentration-baseline comparison above comes from `experiments/legacy/run_value_gap_full_suite.py` and `experiments/legacy/run_phase5_gridworld.py`, and `paper/PROVENANCE.md` maps every reported number to its script and recorded output. All scripts use seeded NumPy generators and are deterministic given the recorded integer seed, and the audit JSON stores the schema version, split sizes, target γ, environment, estimator, and seed in every row.

C Proofs

Every identity and inequality stated in the main text is also checked numerically, independently of the library routines it describes, by `paper/verify_formulas.py`: the two-solve form of $\|\cdot\|_{\mathcal{D}}$ against brute-force enumeration of $M(P)$'s vertices, the seminorm axioms, the regret bound, reparametrization invariance, the zero set $\mathcal{N}$, and the gauge form of the deployed robust objective.

C.1 Proof of Proposition 5

The value function V_θ^π solves the linear Bellman equation $V_\theta^\pi = \Phi_\pi\theta + \beta P_\pi V_\theta^\pi$. Since $\beta \in [0, 1)$ and P_π is stochastic, $\rho(\beta P_\pi) \leq \beta < 1$, so $I - \beta P_\pi$ is invertible via a Neumann series and $V_\theta^\pi = (I - \beta P_\pi)^{-1}\Phi_\pi\theta = W_\pi\theta$. Substituting into the Bellman equation for Q_θ^π,

$$Q_\theta^\pi(s,a) = \phi(s,a)^\top\theta + \beta P(s,a,\cdot)V_\theta^\pi = \phi(s,a)^\top\theta + \beta P(s,a,\cdot)W_\pi\theta = \Psi_\pi(s,a)^\top\theta,$$

which is linear in θ. By policy improvement, π is Bellman-optimal for θ iff $Q_\theta^\pi(s,a) \leq Q_\theta^\pi(s,\pi(s))$ for all s, a; substituting the linear expression yields the stated cone with no relaxation, since the substitution is an equality. The angular calibration score $c_k = \max_{\theta\in\hat{\Theta}_2(\pi_k)} \theta^\top\bar{\theta}$ is then a linear objective over $|S||A|$ linear inequalities plus a unit-ball constraint in $\theta \in \mathbb{R}^d$ alone: an SOCP of size d, independent of $|S|$ once Ψ_{π_k} is precomputed. ■

C.2 Proof of Theorem 2

(a) Define $g_C(x) := \min_{\theta\in C} \theta^\top x$, a pointwise infimum of the affine family $\{x \mapsto \theta^\top x : \theta \in C\}$ and hence concave for any nonempty C, convex or not. Composing with the linear map $\mu \mapsto \Phi^\top\mu$

preserves concavity, so $\mu \mapsto \min_{\theta \in C} \theta^\top \Phi^\top \mu$ is concave in μ. Since $M(P)$ is a fixed polytope not depending on θ, the ambiguity set never touches the feasible region, only the objective, and $\max_{\mu \in M(P)} g_C(\Phi^\top \mu)$ is a concave maximization over a fixed compact convex set. This establishes the convex geometry; efficient solution also requires an efficient representation or linear-minimization oracle for C. No Bellman recursion or rectangularity assumption enters.

(b) Let state s_0 be active and s_1 absorbing with zero reward. At s_0, action a_0 earns zero and moves to s_1; action a_1 earns $1 + t$ and returns to s_0 with probability $1 - t$ (otherwise moving to s_1), where the same ambiguity parameter $t \in [0, 1]$ couples reward and dynamics. Both actions at s_1 self-loop with zero reward. Starting from s_0, let a stationary policy choose a_1 with probability p; its worst-case discounted value is

$$W(p) = \min_{t \in [0,1]} \frac{p(1+t)}{1 - \beta p(1-t)}.$$

At $\beta = 0.9$ we have $W(0) = 0$, $W(5/9) = 10/9$ (constant in t), and $W(5/18) = \min_t 10(1 + t)/(27 + 9t) = 10/27$, attained at $t = 0$. Since $5/18 = \frac{1}{2}(0 + 5/9)$, concavity would require $W(5/18) \geq 5/9$, but $10/27 < 5/9$. Hence W is not concave, and the outer maximization is not a convex program in this regime. ■

C.3 Proof of Proposition 1

Seminorm properties. Nonnegativity is immediate. Positive homogeneity: $V_v(\mu)$ is linear in v, so for $c > 0$, $\max_\mu V_{cv} - \min_\mu V_{cv} = c(\max_\mu V_v - \min_\mu V_v) = c\|v\|_{\mathcal{D}}$. Subadditivity: the supremum of a sum is at most the sum of suprema and the infimum of a sum at least the sum of infima, so $\|u + v\|_{\mathcal{D}} \leq (\max_\mu V_u + \max_\mu V_v) - (\min_\mu V_u + \min_\mu V_v) = \|u\|_{\mathcal{D}} + \|v\|_{\mathcal{D}}$.

Zero set. $\|v\|_{\mathcal{D}} = 0$ iff $V_v(\mu)$ is constant over $M(P)$ iff $v^\top \Phi^\top (\mu - \nu) = 0$ for every $\mu, \nu \in M(P)$, which is the definition of $v \in \mathcal{N}$. Every shaping direction c (via potential Φ_{pot}) satisfies $c^\top \Phi^\top \mu = -(1 - \beta)\rho_0^\top \Phi_{\text{pot}}$ for every $\mu \in M(P)$: expand the flow-conservation constraint $\sum_a \mu(s', a) - \beta \sum_{s,a} P(s, a, s')\mu(s, a) = (1-\beta)\rho_0(s')$ and sum against Φ_{pot}. The constant cancels in $c^\top \Phi^\top (\mu - \nu)$, giving $S_{\text{pot}} \subseteq \mathcal{N}$. The containment can be strict: if a state s is unreachable under every policy then $\mu(s, \cdot) = 0$ throughout $M(P)$, so any v whose induced reward is supported on s lies in $\mathcal{N}$ without being a potential-shaping term. We do not claim strictness in general, only that $\mathcal{N}$ is the exact zero set, which is what the invariance claims use.

Regret bound. With μ^*_θ optimal for θ and $\mu_{\theta'}$ optimal for θ',

$$\begin{aligned}
\text{Reg}_\theta(\pi_{\theta'}) &= V_\theta(\mu^*_\theta) - V_\theta(\mu_{\theta'}) \\
&= V_{\theta - \theta'}(\mu^*_\theta) - V_{\theta-\theta'}(\mu_{\theta'}) + \underbrace{\left[V_{\theta'}(\mu^*_\theta) - V_{\theta'}(\mu_{\theta'})\right]}_{\leq 0 \text{ since } \mu_{\theta'} \text{ maximizes } V_{\theta'}} \\
&\leq \max_\mu V_{\theta-\theta'}(\mu) - \min_\mu V_{\theta-\theta'}(\mu) \;=\; \|\theta - \theta'\|_{\mathcal{D}},
\end{aligned}$$

the last step because $\mu^*_\theta, \mu_{\theta'} \in M(P)$.

Reparametrization invariance. For invertible A, writing $\tilde{\phi} = A\phi$ and $\tilde{v} = A^{-\top} v$, we get $\tilde{\phi}(s, a)^\top \tilde{v} = \phi(s, a)^\top v$ for every (s, a), so the reward function is unchanged, hence $V_v(\mu)$ at every $\mu \in M(P)$, hence $\|v\|_{\mathcal{D}}$. ■

C.4 Proof of Proposition 6

On the quotient by $\mathcal{N}$, $\|\cdot\|_{\mathcal{D}}$ is a norm and every nonnull positive ray has a unit representative, so the triangle inequality, symmetry, and nonnegativity of $d_\circ$ follow from the seminorm, and $d_\circ \leq \|\theta^\circ\|_{\mathcal{D}} + \|\theta'^\circ\|_{\mathcal{D}} = 2$. Independent positive rescaling does not change either unit representative; adding an element of $\mathcal{N}$ changes a representative only by a seminorm-zero vector; feature invariance follows from Proposition 1. If $d_\circ(\theta, \theta') = 0$ then $\theta/\|\theta\|_{\mathcal{D}} - \theta'/\|\theta'\|_{\mathcal{D}} \in \mathcal{N}$, and rearranging gives $\theta' = c\theta + a$ with $c = \|\theta'\|_{\mathcal{D}}/\|\theta\|_{\mathcal{D}} > 0$ and $a \in \mathcal{N}$; the converse is immediate. Finally, optimal policies are unchanged by positive normalization, so applying Proposition 1 to the two normalized rewards gives $\text{Reg}_\theta(\pi_{\theta'})/\|\theta\|_{\mathcal{D}} = \text{Reg}_{\theta^\circ}(\pi_{\theta'^\circ}) \leq \|\theta^\circ - \theta'^\circ\|_{\mathcal{D}}$. ■

C.5 Proof of Proposition 3

Equivalence. By definition $s_k = \min_{\theta \in K(\pi_k)} \|\theta - \bar{\theta}^\circ\|_{\mathcal{D}}$, so $s_k \le q$ iff some $\theta \in K(\pi_k)$ achieves $\|\theta - \bar{\theta}^\circ\|_{\mathcal{D}} \le q$, i.e. iff $K(\pi_k) \cap B_q \neq \emptyset$. The epigraph formulation is a feasible, bounded-below LP over a closed polyhedron, so the optimum is attained; the implementation adds a large numerical safety cap and rejects a solution if that cap is active.

Intersection coverage. $\bar{\theta}$ is a fixed measurable function of the training split, disjoint from the calibration split and the fresh demonstrator. Conditional on $\bar{\theta}$, each s_k is a deterministic measurable function of one i.i.d. demonstrator draw, so $s_1, \dots, s_{N_{\mathrm{val}}}, s_{\mathrm{new}}$ are exchangeable. The ascending rank of s_{new} among these $N_{\mathrm{val}} + 1$ reals is uniform on $\{1, \dots, N_{\mathrm{val}} + 1\}$, so $\mathbb{P}[\mathrm{rank}(s_{\mathrm{new}}) \le \tau] = \tau/(N_{\mathrm{val}}+1) \ge \gamma$ by the choice of τ. With ties broken uniformly at random, $\mathrm{rank}(s_{\mathrm{new}}) \le \tau$ implies $s_{\mathrm{new}} \le q_\gamma$; the converse need not hold with ties, so the nonrandomized threshold is conservative. The sentinel case covers every score. ■

C.6 Proof of Theorem 1

On the Stage-I event, choose a witness $\tilde{\theta} \in K(\pi_{\mathrm{new}})$ with $\|\tilde{\theta} - \bar{\theta}^\circ\|_{\mathcal{D}} \le q_\gamma$. If $q_\gamma < 1$, the reverse triangle inequality gives $\|\tilde{\theta}\|_{\mathcal{D}} \ge 1 - q_\gamma > 0$, and since $K(\pi)$ is a cone, $\tilde{\theta}^\circ$ belongs to $F^\circ(\pi_{\mathrm{new}})$. Moreover

$$\|\tilde{\theta}^\circ - \bar{\theta}^\circ\|_{\mathcal{D}} \le \|\tilde{\theta}^\circ - \tilde{\theta}\|_{\mathcal{D}} + \|\tilde{\theta} - \bar{\theta}^\circ\|_{\mathcal{D}} \le |1 - \|\tilde{\theta}\|_{\mathcal{D}}| + q_\gamma \le 2q_\gamma.$$

The latent normalized reward lies in the same cone, so Assumption 2 and the triangle inequality give $\|\hat{\theta}^\circ_{\mathrm{new}} - \bar{\theta}^\circ\|_{\mathcal{D}} \le 2q_\gamma + \eta$; when $q_\gamma \ge 1$ this is subsumed by the cap $R = 2$ from the universal projective diameter. Proposition 3 assigns probability at least γ to the Stage-I event.

For the robust statement, μ_{ref} is feasible and has relative value zero under every reward, so the optimum at μ_R is nonnegative. On the containment event $\hat{\theta}^\circ_{\mathrm{new}} \in B_R$, hence its relative value at μ_R is at least the worst-case relative value and therefore at least zero; positive rescaling back to $\hat{\theta}_{\mathrm{new}}$ preserves the inequality. Subtracting both values from the same optimal value gives the equivalent regret statement. ■

C.7 Proof of Proposition 4

Let $x = \Phi^\top(\mu - \mu_{\mathrm{ref}})/(1-\beta)$. Polar duality and $\|\bar{\theta}^\circ\|_{\mathcal{D}} = 1$ give $V_{\bar{\theta}^\circ}(\mu) - V_{\bar{\theta}^\circ}(\mu_{\mathrm{ref}}) \le \gamma_{D_F}(x)$, so the relative robust objective is at most $(1-R)\gamma_{D_F}(x) \le 0$ when $R \ge 1$. The reference has $x = 0$ and attains zero. If $R > 1$, equality requires $\gamma_{D_F}(x) = 0$, i.e. the two occupancies have the same feature vector modulo the null directions; at $R = 1$ additional exposed-face ties can attain zero. ■

D Post-acceptance supplement: an exact cone diagnostic

Added 6 October 2026, after acceptance; this appendix was not part of the reviewed submission. This small construction exercises the containment bridge with a computed certificate and separates its conservatism from that of a ball-shaped uncertainty set. It changes none of E1–E3 or Table 2.

An exactly solvable geometry. There is one self-looping state, discount $\beta = 0.9$, and m actions with features $p_j = (\cos(2\pi j/m), \sin(2\pi j/m))$, $j = 0, \dots, m-1$; $m \in \{4, 16\}$. Occupancies are action distributions and $V_\theta(\mu) = \sum_j \mu_j p_j^\top \theta/(1-\beta)$. Central symmetry gives

$$\|\theta\|_{\mathcal{D}} = \frac{2\max_j p_j^\top \theta}{1-\beta}.$$

Write $h = \pi/m$. The Bellman cone for action j is the angular sector $[2jh - h, 2jh + h]$. Inside it, p_j attains the maximum above. Consequently $F^\circ(j)$ is exactly the line segment with endpoints

$$u_j^\pm = \frac{1-\beta}{2\cos h}\big(\cos(2jh \pm h), \sin(2jh \pm h)\big).$$

Indeed, intersecting the sector with $p_j^\top \theta = (1-\beta)/2$ gives precisely that segment. Thus, for any fixed center, convexity of the norm yields

$$s^{\max}(j) = \max_{\sigma \in \{-,+\}} \|u_j^\sigma - \bar{\theta}^\circ\|_{\mathcal{D}}, \qquad \mathrm{diam}_{\mathcal{D}}\, F^\circ(j) = \|u_j^+ - u_j^-\|_{\mathcal{D}} = 2\tan h.$$

For the last equality, $u_j^+ - u_j^- = (1-\beta)\tan(h)\, p_j^\perp$, and $p_j^\perp$ is an action feature because m is divisible by four. This is a certificate for *every* action's entire normalized cone, obtained from the known MDP, not a maximum over sampled latent rewards. The example needs no external side information. Endpoint maxima suffice here because each normalized cone is a segment; we claim no general efficient solver for $s^{\max}$.

Protocol and independent checks. Fix the unit-span center $\bar{\theta}^\circ = ((1-\beta)/2, 0)$ and the uniform reference $\mu_{\mathrm{ref},j} = 1/m$ before drawing any calibration or test data. This isolates the certificate and is not a comparison of IRL estimators. For $m = 16$, draw the optimal action J from $\{-1, 0, 1\}$ modulo m with probabilities $(1/4, 1/2, 1/4)$. Draw U uniformly on $[-0.45, 0.45]$ and span-normalize the reward direction at angle $2\pi(J+U)/m$. These rewards have unique optimal action J. For the $m = 4$ control, use $J = 0$ always and the same continuous within-cone distribution. Use 100 independent trials, each with 20 calibration rewards and 100 independent test rewards, target $\gamma = 0.8$, and master seed 20261006. Calibration and test draws use separate random streams. All reported empirical rates average trial-level rates; test points sharing a calibrated radius are not treated as independent trials.

Compute three calibration quantiles on the same split: q_γ from the minimum-cone LP score; $q_\gamma^{\max}$ from the exact endpoint maximum; and the diagnostic oracle $q_\gamma^{\mathrm{oracle}}$ from the distances of the *latent calibration rewards*. Only the oracle uses latent calibration rewards; no method uses test rewards to set a radius. Pointwise ordering gives

$$q_\gamma^{\mathrm{oracle}} \le q_\gamma^{\max} \le R = \min\{2, 2q_\gamma + \eta\}.$$

The oracle is not a deployable IRL method. The script checks the analytic cone endpoints and compares the library's cone score with an independent SciPy LP. It also audits each deployed policy's worst-case value with an independent two-variable inner LP. Tests compare the endpoint support values with LPs over all Bellman inequalities and the exact span-normalization equality.

A positive certificate. For $m = 16$, $\eta = 2\tan(\pi/16) = 0.397825$. All 100 calibration splits give the same q_γ, $q_\gamma^{\max}$ and R, up to numerical precision (Table 3). Both robust programs select action 0 to solver tolerance, whose reference-relative worst-case value over a ball of radius $r < 1$ is $(1-r)/2$. To see optimality, central symmetry implies $D_F = 2\,\mathrm{conv}\{p_j\}/(1-\beta)$, so the gauge of the feature difference for action 0 is $1/2$. Every action distribution has gauge at most $1/2$ and center value at most that gauge, giving the upper bound $(1-r)/2$, attained by action 0. Thus $R < 1$ is accompanied here by an independently verified *positive* worst-case advantage and a departure from the reference.

Table 3: Post-acceptance diagnostic. J_R and $J_{\max}$ are worst-case advantages over the uniform reference using radii R and $q_\gamma^{\max}$, respectively. The displayed radii and values are common to all 100 splits to the shown precision; they are not performance comparisons with a point estimator.

m	η	q_γ	$q_\gamma^{\max}$	R	J_R	$J_{\max}$
16	0.397825	0.191342	0.581596	0.780508	0.109746	0.209202
4	2	0	1	2	0	0

For $m = 16$, Stage-I containment is 0.5039; exact-maximum and bridge containment are both 1.0000 on the 10,000 held-out draws. Oracle containment is 0.8173 (standard error 0.0087 across 100 independent trial means); its mean radius is 0.428040. The endpoint score is discrete and the population lies inside its cones, so its conservative 1.0000 containment should not be read as evidence of tight calibration. Reference dominance holds on all test draws. These rates are descriptive checks; the certificate is the analytic cone bound and the theorem, not observed test coverage. Both robust radii select the same action, so the larger $J_{\max}$ is a tighter certificate, not a measured policy-performance gain.

A set-shape control. For $m = 4$, the observed action is always 0, $q_\gamma = 0$, and the full normalized cone has diameter 2. Its exact maximum-distance score is 1, so even exact-maximum calibration reaches the ball's zero-advantage threshold. The implementation chooses the reference at that threshold; at $r = 1$ other optimizers can tie, whereas at $R = 2$ all optimizers are reference-feature-equivalent. Nevertheless, optimizing directly over $F^\circ(0)$ gives worst-case advantage $1/2$: every reward in that segment gives action 0 value $1/2$, and the uniform reference has value zero. A

two-endpoint robust LP verifies this value without observing latent rewards. This diagnostic uses the stated population of action-0 demonstrators; it is not a distribution-free replacement for the population-calibrated method. It shows that a wide reward cone and a degenerate enclosing ball do not imply that all safe decision rules are uninformative.

Reproduction and limits. `make toy` runs `experiments/run_exact_cone_toy.py`; configuration, analytic endpoints, scores, occupancies and every trial are stored in `experiments/exact_cone_toy_results.json`. This is a one-state, two-feature existence example with a fixed center and reference. It neither certifies the widths in E1–E3 nor establishes scalability, improvement over point-estimate IRL, or robustness to estimated policies or dynamics.